

Question Paper

Exam Date & Time: 22-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Parallel Computer Architecture [NPT_4132]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Explain how Flynn's taxonomy can be extended to classify modern heterogeneous architectures. (5)
Discuss its limitations in today's context
- A) Course Outcome: "CO1" / Cognitive Level: "Application"
- B) Analyze the trade-offs between UMA and NUMA architectures in large-scale multiprocessors. How do these choices affect memory latency and programmability? (5)
- Course Outcome: "CO1" / Cognitive Level: "Analysis"
- 2) Demonstrate how the message-passing model can be used to compute the matrix-vector product on a 4-node cluster. Show the data distribution and communication pattern. (5)
- A) Course Outcome: "CO2" / Cognitive Level: "Evaluate"
- B) Analyze the role of the VI protocol as a baseline coherence mechanism. How does its simplicity help in reasoning about correctness compared to more complex protocols like MESI? (5)
- Course Outcome: "CO2" / Cognitive Level: "Evaluate"
- 3) Assess how coherence correctness like write serialization, data consistency, and atomicity is maintained in Dragon and MESI protocols. Discuss failure scenarios that could violate correctness. (5)
- A) Course Outcome: "CO2" / Cognitive Level: "Analysis"
- B) Develop an evaluation framework to measure coherence performance in a multicore system with single vs. multi-level caches. Identify metrics that reflect both latency and traffic efficiency (5)
- Course Outcome: "CO3" / Cognitive Level: "Analysis"
- 4) Compare directory-based coherence to snooping protocols for large-scale multiprocessors in terms of scalability, fault tolerance, and hardware complexity. Provide design reasoning for preferring one in NUMA systems. (5)
- A) Course Outcome: "CO3" / Cognitive Level: "Analysis"
- B) Evaluate the Sequent NUMA-Q architecture's coherence protocol for correctness and latency. Propose a hybrid coherence approach combining snooping and directory methods for improved performance. (5)

Course Outcome: "CO3" / Cognitive Level: "Evaluate"

- 5) Analyze the trade-offs between strict consistency and release consistency in parallel systems. (5)
Discuss how synchronization primitives help enforce ordering under relaxed models.

A)

Course Outcome: "CO4" / Cognitive Level: "Evaluate"

- B) Evaluate the performance implications of point-to-point synchronization versus global barriers in (5)
NUMA architectures. Suggest design improvements to reduce synchronization overhead.?

Course Outcome: "CO4" / Cognitive Level: "Analysis"

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Event Name	Subject
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058-END SEMSTER-ODD-2025 NPT_4132:Parallel Computer Architecture

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	15
3	3	CO 3	15
4	4	CO 4	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025 NPT_4132:Parallel Computer Architecture

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	1	5(10%)
4	Analysis	5	25(50%)
5	Create	4	20(40%)
6	Evaluate	0	0(0%)
Total		10	50 (100%)